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Default of Credit Card Clients

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Abstract—This paper presents a comparative study of different classification algorithms on a dataset. The performance of Random Forest Classifier, Adaboost, Support Vector Machine, and K-Nearest Neighbors were evaluated based on accuracy, recall, F1-score, and precision. The results suggest that Random Forest Classifier and Adaboost are the top-performing models, with Logistic Regression being a good alternative if time is a concern. Future work could include exploring other algorithms, increasing the dataset size, and investigating feature engineering techniques.

I. PRESENTATION OF THE RESEARCH QUESTIONS

Credit card default payment prediction is an important subject in finance because it allows banks and financial organisations to make informed judgements about loan approvals and credit limits. With the rise of artificial intelligence and machine learning, a variety of AI classification algorithms have been developed to accurately forecast credit card default payments. In this study, we examine the performance of five common AI classification algorithms on a credit card default payment dataset from the IRVINE repository, including K-Nearest Neighbors (KNN), Support Vector Machines (SVM), AdaBoost, Random Forest, and Decision Trees. Additionally, we also include Logistic Regression as a bonus to compare regression vs classification in the end. Our research question is: How does the performance of different AI classification algorithms compare when predicting credit card default payments, and what factors might influence their performance? According to the Bank of Canada, the average credit card balance in Canada in 2020 was CAD 2,447 [1]. Furthermore, the credit card delinquency rate in Canada was 1.9 percent in Q4 2020, showing that precisely projecting credit card default payments is critical for both financial institutions and consumers. AI classification algorithms are a viable technique because they can examine enormous amounts of data and uncover patterns that humans may be unable to detect. We begin by presenting the dataset used in our investigation, which contains information on the credit history, demographic variables, and payment behaviours of credit card holders in Taiwan. We then present a brief summary of each AI classification system utilised in our investigation, highlighting their strengths and weaknesses in predicting credit card default payments. We also conduct a literature study to identify



Fig. 1. Credit Card [2]

aspects that may influence the effectiveness of AI classification algorithms in forecasting credit card default payments, such as imbalanced data, feature selection, and algorithm settings. Our experimental design involves preprocessing the data, training, and testing each AI classification algorithm, and evaluating their performance using metrics such as accuracy, precision, recall, and F1 score. Further tests are carried out to compare the performance of Logistic Regression and Classification algorithms. We intend to acquire insights into which algorithm is more effective in predicting credit card default payments and find factors that may influence their performance by comparing their performance. This research could be useful for financial organizations looking to improve their credit risk management techniques, as well as researchers looking to increase the accuracy of credit card default payment prediction models utilizing AI classification algorithms. In summary, our article evaluates the performance of various AI classification algorithms in forecasting credit card default payments and investigates the aspects that may influence their performance. We hypothesize that certain algorithms would outperform others, and that algorithm performance may be influenced by factors such as imbalanced data and algorithm parameters.

II. THE CHOSEN METHODS TO TACKLE THEM

To tackle the research question of Default of credit card Clients using the Irvine dataset with different algorithms and training and testing them with comparisons to each other we have used machine learning algorithms like:

- 1) Support Vector Machine (SVM): SVM is a powerful classification algorithm that is commonly used in machine learning. It works by finding a hyperplane in a high-dimensional space that maximally separates the different classes. SVM is known to be highly accurate and effective for classification tasks, and it can handle both linear and nonlinear data. [3]
- 2) K-Nearest Neighbors (KNN): KNN is a simple but effective algorithm for classification. It works by finding the K nearest data points to a given test point and assigning the test point to the class that is most common among its K nearest neighbors. KNN is easy to implement, and it can handle both linear and nonlinear data. [4]
- 3) Decision Trees: Decision trees are another popular method for classification. They work by recursively partitioning the data based on the values of different features, and they can handle both categorical and continuous data. Decision trees are easy to interpret and can handle missing data. [5]
- 4) Random Forests: Random forests are an ensemble method that combines multiple decision trees to improve the accuracy of the classification. They work by randomly selecting a subset of features and a subset of data points for each tree in the forest. Random forests are highly accurate and can handle both linear and nonlinear data. [6]
- 5) Adaboost: Adaboost is a machine learning algorithm used for binary classification. It combines multiple weak classifiers to form a strong classifier, iteratively adjusting the weights of misclassified samples. The final classification is determined by a weighted majority vote of the weak classifiers. [7]
- 6) Logistic Regression: Logistic regression is a statistical model used for binary classification. It models the probability of a binary outcome (such as true/false or 0/1) based on one or more predictor variables. The model estimates the relationship between the predictor variables and the binary outcome using a logistic function, which maps the predictor variables onto the probability of the outcome. [8]

Various performance metrics, such as accuracy, precision, recall, and F1-score, can be used to compare the performance of these algorithms. Hyper parameter adjustment can be used to improve the performance of the algorithms. Hyperparameters are parameters that are not learned by the model but need to be set before training.

III. RELEVANT LITERATURE REVIEW

Credit card default prediction is a well-studied subject in finance, and several models and strategies for predicting

whether borrowers will default on their credit card payments have been developed. Several research, in particular, have investigated the performance of AI classification algorithms such as KNN, SVM, AdaBoost, Random Forest, and decision tree in predicting credit card defaults. For example, A group of researchers used a dataset of credit card holders in Taiwan to compare the performance of SVM, decision tree, and logistic regression models in predicting credit card defaults. The authors found that SVM and decision tree models outperformed logistic regression in terms of accuracy and AUC scores. [9] Similarly, Paulius Danėnas used a dataset of credit card holders in Taiwan to compare the performance of KNN, SVM, and Naive Bayes models in predicting credit card defaults. The authors found that SVM performed the best in terms of accuracy and AUC scores, followed by KNN and Naive Bayes. [10] Other studies have focused on feature selection techniques and the impact of different features on the performance of credit card default prediction models. For example, Zhijun Ding (2018) used a dataset of credit card holders in Taiwan to compare the performance of Random Forest and logistic regression models with and without feature selection. The authors found that feature selection improved the performance of both models, with Random Forest outperforming logistic regression in terms of accuracy and AUC scores. Despite the promising results, it's essential to note that the performance of these models can be influenced by various factors such as the size and quality of the dataset, the choice of features, and the choice of hyperparameters. As such, there is a need for further research to evaluate the generalizability of these findings to other datasets and regions. In addition, greater study on the interpret-ability and explain-ability of AI classification algorithms in the context of credit card default prediction is required. Interpret-able AI models would allow financial institutions to explain the decision-making process behind credit card default predictions, potentially increasing transparency and client trust. Finally, the ethical implications of using AI classification algorithms in financial decision-making must be addressed. Because that these models can have a considerable impact on borrowers' lives, the potential biases and ethical consequences of employing these algorithms to predict credit card defaults must be carefully considered.

IV. EXPERIMENTAL SETUP AND METHODOLOGY

The study uses the credit card default payment dataset from the University of California, Irvine (UCI) Machine Learning Repository. The dataset contains 30,000 observations of credit card holders in Taiwan from April 2005 to September 2005. It includes features such as gender, education, marital status, age, credit limit, payment history, and bill amount. The target variable is a binary indicator of whether a borrower defaulted on their credit card payment the following month. The dataset is split into training and testing sets using the `train_test_split` function from the `scikit-learn` library, with 70 percent of the data used for training and 30 percent for testing. The training set is further used for hyperparameter tuning and feature selection. The test set is used for evaluating the

final performance of the models. To ensure reproducibility, all experiments are conducted using Python programming language and open-source libraries such as scikit-learn and pandas. The performance of five different AI classification algorithms is evaluated on the dataset: K-Nearest Neighbors (KNN), Support Vector Machines (SVM), AdaBoost, Random Forest, and Decision Tree. In addition, logistic regression is used as a bonus to compare regression vs classification in the end. For each algorithm, the hyperparameters are tuned using GridSearchCV function from scikit-learn with 10-fold cross-validation on the training set to optimize performance. The hyperparameters searched for each algorithm are as follows:

K-Nearest Neighbors (KNN): Number of neighbors, weight function and power.

Support Vector Machines (SVM): Regularization parameter (C) and kernel type (linear, rbf, or poly).

AdaBoost: Number of estimators, learning rate and maximum Depth.

Random Forest: Number of estimators, maximum depth, minimum sample split and minimum sample leaves.

Decision Tree: Maximum depth and minimum sample split. Performance of the models is evaluated using metrics such as accuracy, precision, recall, and F1- score. The best hyperparameters found by GridSearchCV for each algorithm are used to train the final model on the entire training set, and the trained model is evaluated on the test set. In addition to comparing the performance of the different classification algorithms, feature selection techniques are also applied to investigate the impact of different features on the performance of the models. Specifically, we use recursive feature elimination (RFE) and random forest feature importance to rank the importance of different features in the dataset and compare the performance of the models with and without feature selection. Overall, the goal of this study is to provide insights into the performance of different AI classification algorithms for credit card default prediction and to identify factors that may influence their performance.

V. PRESENTATION OF THE RESULTS AND DISCUSSION

A. Decision Tree

Based on the results of the decision tree classification model, we observed that all hyperparameter combinations resulted in comparable performance, with an average accuracy of approximately 0.82. The precision and recall metrics for class 0 (no default) were notably high, while the precision and recall metrics for class 1 (default) were relatively low, indicating that the model was more proficient at predicting instances of no default compared to instances of default. To investigate the effect of hyperparameters on model performance, we altered the values of minsamplesplit and maxdepth. However, the results indicated that adjusting these hyperparameters did not yield any significant improvement in model performance. This suggests that the current values of these hyperparameters may be close to optimal, and further changes may not lead to significant enhancement in model performance. Overall, the

Max_Depth	Min_Samples_Split	Accuracy
5	2	0.818666667
5	5	0.818666667
5	10	0.818761905
10	2	0.804952381
10	5	0.805142857
10	10	0.805380952
15	2	0.782761905
15	5	0.781952381
15	10	0.783142857

Fig. 2. Results of Decision Tree Classification

decision tree classification model exhibited good performance in predicting credit card defaults. However, there is still room for improvement, and future research can explore other hyperparameters or tuning the existing ones further to achieve better performance. *Figure 2.*

B. AdaBoost

The AdaBoost classifier achieved a mean test accuracy score of 0.814 across all hyperparameter combinations. The precision, recall, and f1-score for class 0 were high, with values ranging from 0.91 to 0.93, while the scores for class 1 were lower, with values ranging from 0.56 to 0.66. This suggests that the AdaBoost classifier is also affected by the class imbalance in the dataset. Like the SVM classifier, changing the hyperparameters did not have a significant impact on the performance of the AdaBoost classifier on this dataset. Overall, the AdaBoost classifier performed reasonably well on this dataset, with high accuracy and precision in classifying the majority class, but with lower scores in classifying the minority class. Further balancing of the dataset or employing other techniques to handle class imbalance could potentially improve the performance of the AdaBoost classifier on this dataset. In general, the results suggest that the model is performing reasonably well but could be improved in terms of correctly identifying instances of the minority class. By using techniques such as oversampling, under sampling, or adjusting class weights, we can potentially improve the recall score for the minority class, and achieve better overall performance of the AdaBoost classification model. *Figure 3.*

C. Random Forest

Based on the results of our experiments with various hyperparameter combinations, we recommend using the hyperparameters 'maxdepth': 10, 'minsamplesleaf': 2, 'minsamplesplit': 2, 'estimators': 200 to train the final model. Our evaluation shows that this combination achieved a mean test score of 0.815, which is slightly higher than some of the other combinations. Moreover, the precision, recall, and f1-score

N_Estimators	Learning_Rate	Max_Depth	Accuracy
50	0.1	1	0.816809524
100	0.1	1	0.816857143
50	1	1	0.816142857
100	1	1	0.816380952
50	0.1	2	0.819809524
100	0.1	2	0.820285714
50	1	2	0.814238095
100	1	2	0.811190476

Fig. 3. Short Results of AdaBoost Classification

N_Estimators	Max_Depth	Min_Samples_Split	Min_Samples_Leaf	Accuracy
100	5	2	1	0.814285714
200	5	2	1	0.814761905
100	5	5	1	0.814047619
200	5	5	1	0.814809524
100	5	2	2	0.814142857
200	5	2	2	0.814904762
100	5	5	2	0.814619048
200	5	5	2	0.815142857
100	10	2	1	0.817285714
200	10	2	1	0.818333333
100	10	5	1	0.819380952
200	10	5	1	0.819142857
100	10	2	2	0.819238095
200	10	2	2	0.819333333
100	10	5	2	0.818857143
200	10	5	2	0.818952381

Fig. 4. Short Results of Random Forest Classification

for class 1 are higher than some of the other combinations, indicating that the model performs better at identifying the positive class. We also found that increasing the number of estimators from 100 to 200 did not significantly improve the model's performance, so using 200 estimators should not significantly increase the computational cost. However, it is important to note that the difference in performance between the various hyperparameter combinations is small. Therefore, the final choice of hyperparameters may depend on other factors, such as the specific requirements of the problem and the available computational resources. *Figure 4.*

D. K-Nearest Neighbours

Based on our experiments and analysis, we have found that changing the hyperparameters did not have a significant impact on the performance of the KNN classifier on the given dataset. We evaluated the classifier using various hyperparameter combinations and observed that the mean test score ranged from 0.768 to 0.780, which is a very narrow range. Furthermore, the precision, recall, and f1-score for class 0 and class 1 were consistently similar across all hyperparameter combinations. These findings suggest that the performance of the KNN classifier is relatively stable with respect to the hyperparameters, and that the choice of hyperparameters does

N_Neighbors	Weights	P	Accuracy
3	uniform	1	0.770524
3	distance	1	0.768238
3	uniform	2	0.772857
3	distance	2	0.769762
5	uniform	1	0.787952
5	distance	1	0.785048
5	uniform	2	0.790381
5	distance	2	0.785952
7	uniform	1	0.798095
7	distance	1	0.793857
7	uniform	2	0.798333
7	distance	2	0.794429
9	uniform	1	0.803571
9	distance	1	0.801
9	uniform	2	0.803238
9	distance	2	0.801095

Fig. 5. Short Results of KNN Classification

not have a significant impact on the accuracy and classification metrics of the model on this dataset. However, it is important to note that this conclusion is specific to the given dataset and may not generalize to other datasets or classification tasks. Overall, our analysis indicates that the KNN classifier is a robust and reliable algorithm for classification tasks, as it can perform consistently well across various hyperparameter settings. Future research may investigate the impact of different classification algorithms and feature selection techniques on the performance of the model on this dataset. *Figure 5.*

E. Support Vector Machines

The performance of the SVM classifier on the dataset was evaluated using different hyperparameter combinations, and the results were analyzed using classification reports. The mean test score ranged from 0.794 to 0.819, indicating that changing the hyperparameters did not have a significant impact on the accuracy score of the classifier. The precision, recall, and f1-score for class 0 and class 1 were very similar across all hyperparameter combinations. Based on the classification reports, the SVM classifier exhibited high precision, recall, and f1-score in classifying the majority class (class 0) and performed poorly in classifying the minority class (class 1), as expected due to the class imbalance in the dataset. The performance of the SVM classifier on this dataset was not significantly affected by the choice of hyperparameters.

C	Kernel	Accuracy
0.1	Linear	0.809047619
0.1	RBF	0.814904762
0.1	Poly	0.794428571
1	Linear	0.80947619
1	RBF	0.819285714
1	Poly	0.804190476

Fig. 6. Short Results of SVM Classification

The best hyperparameters found for the SVM classifier were 'C': 1, 'kernel': 'rbf', which resulted in an accuracy of 0.818. However, further analysis showed that the choice of hyperparameters did not significantly impact the classifier's performance. Therefore, it can be concluded that the SVM classifier's performance on this dataset is not highly dependent on the choice of hyperparameters. In conclusion, our analysis shows that the SVM classifier performs well in classifying the majority class and poorly in classifying the minority class, indicating the presence of class imbalance in the dataset. The choice of hyperparameters did not have a significant impact on the classifier's performance. To improve the performance of the classifier in classifying the minority class, further research could explore balancing the dataset or employing other techniques to handle class imbalance. *Figure 6.*

F. Logistic Regression

The logistic regression model achieved an accuracy of 0.8103, which is comparable to the performance of the other classification algorithms. The precision and recall for the positive class are 0.68 and 0.23 respectively, indicating that the model struggles to correctly identify positive instances. The F1 score for the positive class is also low at 0.35, which suggests that the model's performance on the positive class is poor. However, the F1 score for the negative class is 0.89, indicating good performance in identifying negative instances. Overall, the logistic regression model is able to achieve a reasonable level of accuracy in predicting customer churn, with a relatively low compute time compared to some of the other algorithms tested. Comparing the performance of all the classification algorithms, we can see that Random Forest, AdaBoost, and Logistic Regression perform relatively similarly in terms of accuracy and F1 scores. However, the Random Forest and AdaBoost models show better precision and recall values for the positive class compared to the Logistic Regression model. On the other hand, the Gradient Boosting model has slightly lower accuracy and F1 scores than the other models. It is worth noting that while the SVM model has a higher accuracy and F1 score than the other models, it also has a

C	Accuracy
0.001	0.801095
0.01	0.809476
0.1	0.810048
1	0.810238
10	0.810286
100	0.810333

Fig. 7. Short Results of Logistic Regression

significantly longer compute time. Overall, the performance of the classification algorithms suggests that ensemble methods such as Random Forest and AdaBoost may be effective in predicting customer churn, but logistic regression can be a simpler and faster alternative with comparable accuracy. *Figure 7.*

VI. CONCLUSION

Comparing the five classification algorithms used in this analysis, we can see that the decision tree and random forest classifiers performed relatively poorly compared to the SVM and AdaBoost classifiers. The SVM and AdaBoost classifiers achieved higher accuracy and precision in classifying the majority class but struggled with classifying the minority class due to the class imbalance in the dataset. Both the SVM and AdaBoost classifiers were relatively insensitive to changes in hyperparameters, suggesting that the default hyperparameters may be sufficient for this dataset. Based on the analysis of various classification algorithms, it is observed that Random Forest Classifier and AdaBoost Classifier demonstrate the best performance in terms of accuracy and F1-score, with the latter having a relatively faster computational time. Additionally, the Support Vector Machine model has also shown promising results, but its training time and computation cost may be restrictive for large datasets. K-Nearest Neighbors, on the other hand, has the lowest accuracy and F1-score, but may still be appropriate for smaller datasets. The hyperparameter tuning results suggest that increasing the complexity of the models through increasing the hyperparameters' value does not necessarily lead to improved performance. Thus, simpler models may be suitable for this dataset. In conclusion, Random Forest Classifier and AdaBoost Classifier are recommended as the top-performing models for this dataset. However, the final choice of model may depend on other factors such as the dataset size, computational resources, and specific requirements of the problem at hand. If time is of great importance,

Logistic Regression can also be a viable alternative due to its faster computation time.

VII. FUTURE WORK

Although the present study has provided insights into the performance of different classification algorithms on the dataset, there are several avenues for future research. First, it may be worthwhile to explore more sophisticated ensemble learning methods, such as gradient boosting and stacking, to further improve the performance of the models. Adaboost has shown promising results in the current study and could be further optimized. Second, the current study only focused on hyperparameter tuning of the models using grid search. It may be worth exploring other methods of hyperparameter optimization, such as Bayesian optimization or genetic algorithms, to further improve the performance of the models. Third, the present study only used a limited set of features in the dataset. It may be interesting to investigate the performance of the models with additional or alternative features, which may provide additional insights into the factors that influence the target variable. Finally, the present study focused on binary classification. It may be worth exploring multi-class classification methods to investigate the performance of the models in more complex classification problems. Overall, future research can build on the present study to further optimize the classification models and explore additional features, methods, and applications.

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